

De-immunized Protein Design via a Unified, Verified MHC-II Immunopeptidome Dataset

– Dataset description

a) Overview

We have assembled MHC Class II Immunopeptidome Dataset (Unified & Verified), a large-scale, quality-controlled dataset consolidating experimentally observed MHC-II ligands from two major public sources

IEDB (Low Resolution): 740,383 records

IEDB (Standard Resolution): 364,025 records

HLA Ligand Atlas: 142,464 records

Total: 1,246,872 peptide–protein records (TSV; primary file: mhc_if_v2.tsv)

Crucially, every record is cross-verified against UniProt protein sequences with 100% sequence–coordinate consistency, enabling robust mapping from peptide-level immunopeptidome evidence to full-length protein sequence positions.

Each row links an observed ligand peptide to its source protein and includes:

Peptide sequence (peptide_seq)

MHC class fixed as Class II (hla_class = II)

Alleles (alleles, JSON array), with an explicit allele-resolution label (resolution) to distinguish generic/serotype/high-res single/dimer/mixed

Source tissue (tissues, JSON array) when available

Source protein accessions (protein_accessions, JSON array)

Verified positional grounding (peptide_position_info, JSON array): 1-based start/end coordinates (IEDB convention)

Local context (flanking_context, JSON array): 3-mer upstream/downstream residues around the peptide

b) Processing, QC, and provenance

The dataset was generated via a rigorous pipeline:

1. Coordinate extraction from raw IEDB exports
2. Position discovery for Atlas peptides via substring search over UniProt sequences
3. Mapping all proteins to UniProt accessions and downloading >16,000 full-length sequences (FASTA available)
4. Automated correction of minor coordinate shifts (~7.8k cases)
5. Flank extraction directly from verified sequences
6. A final global verification ensuring:

Peptide Sequence == Protein Sequence[start:end] for every retained record

Rows failing verification were removed; if a peptide occurs multiple times in a protein, all valid matches are retained.

This yields a combination: scale + strict verification + positional context, which is essential for building reliable full-length immunogenicity models.

– Research objectives

Motivation

Current epitope/ligand predictors are typically trained and evaluated on peptide-only inputs, which limits their utility for protein engineering, where practitioners need to (i) identify high-risk regions across a full protein, and (ii) redesign sequences under structural/function constraints. Meanwhile, state-of-the-art inverse folding tools (e.g., ProteinMPNN, diffusion-based inverse folding) optimize for foldability but are not immunogenicity-aware, often leaving strong MHC-II hotspots untouched, causing severe immune response – antidrug antibody.

Our goal is to build the first modular framework that injects MHC-II immunopeptidome constraints into protein inverse folding/design, enabled by our verified peptide-to-protein mapping dataset.

Aim 1: Full-length hotspot prediction from peptide-level evidence

Objective: Train a sequence-level “epitope-risk head” that takes a full-length protein sequence as input and outputs a position-wise risk map (hotspots) and an aggregated global risk score.

Aim 2: Hotspot-aware protein design via diffusion guidance

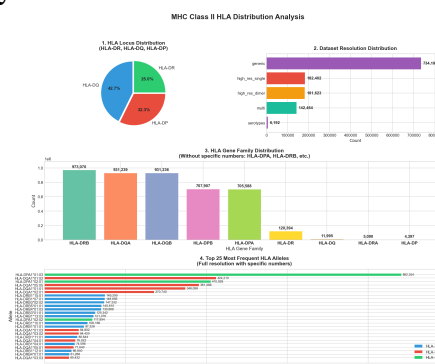
Objective: Integrate the trained risk head as a guidance module for inverse folding / generative design (diffusion-based inverse folding preferred), producing sequences that reduce predicted MHC-II hotspots while maintaining structural plausibility.

Aim 3: Modularization: plug-and-play immune constraint for the community

Objective: Release the epitope-risk head as a modular component that can be attached to different protein generative models (inverse folding, sequence diffusion, language models), enabling broad community use.

– Preliminary analyses

Data distribution analyses:



Overlap analysis:

	Number	Proportion
Unique peptides	412,158	-
Single source	271,345	65.84%
Overlap	140,813	34.16%

But if we consider pair overlap (peptide, allele):

	Number	Proportion
Unique pairs	708,006	-
Single source	707,544	99.93%
Overlap	462	0.07%